Homework 03

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1 Problem 3.1 (Risk-Neutral SLP with Recourse)

1.1 (a)

The recourse problem can be formulated as the following:

$$
\begin{aligned}
\text{max} \quad & \sum_{t \in T} \left(P_t^{DA} x_{it} + \mathbb{E}\left[P_t^{RT}(\xi) y_{it}^{+}(\xi) - P_t^{PN} y_{it}^{-}(\xi) \right] \right) && \text{(1a)} \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) \quad \forall t \in T && \text{(1b)} \\
& R_{it}(\xi) \geq y_{it}^{+}(\xi) \quad \forall t \in T && \text{(1c)} \\
& y_{it}^{+}(\xi) \leq M z_{it}(\xi), \quad y_{it}^{-}(\xi) \leq M(1 - z_{it}(\xi)) \quad \forall t \in T && \text{(1d)} \\
& x_{it}^{DA} \geq 0, y_{it}^{+}(\xi) \geq 0, y_{it}^{-}(\xi) \geq 0, z_{it}(\xi) \in \{0, 1\} \quad \forall t \in T && \text{(1e)}
\end{aligned}
$$

1.2 (b)

2 Problem 3.2 (Deterministic Models)

2.1 (a)

2.2 (b)

2.3 (c)

2.4 (d)

3 Problem 3.3 (Excess Probability)

4 Problem 3.4 (Conditional Value-at-Risk)

5 Problem 3.5 (Expected Excess)

6 Problem 3.6 (Probabilistically Constrained SP)

7 Problem 3.7 (Quantile Deviation)

8 Problem 3.11 (Risk Measure)